SERVICE MANAGEMENT IN CLOUD COMPUTING

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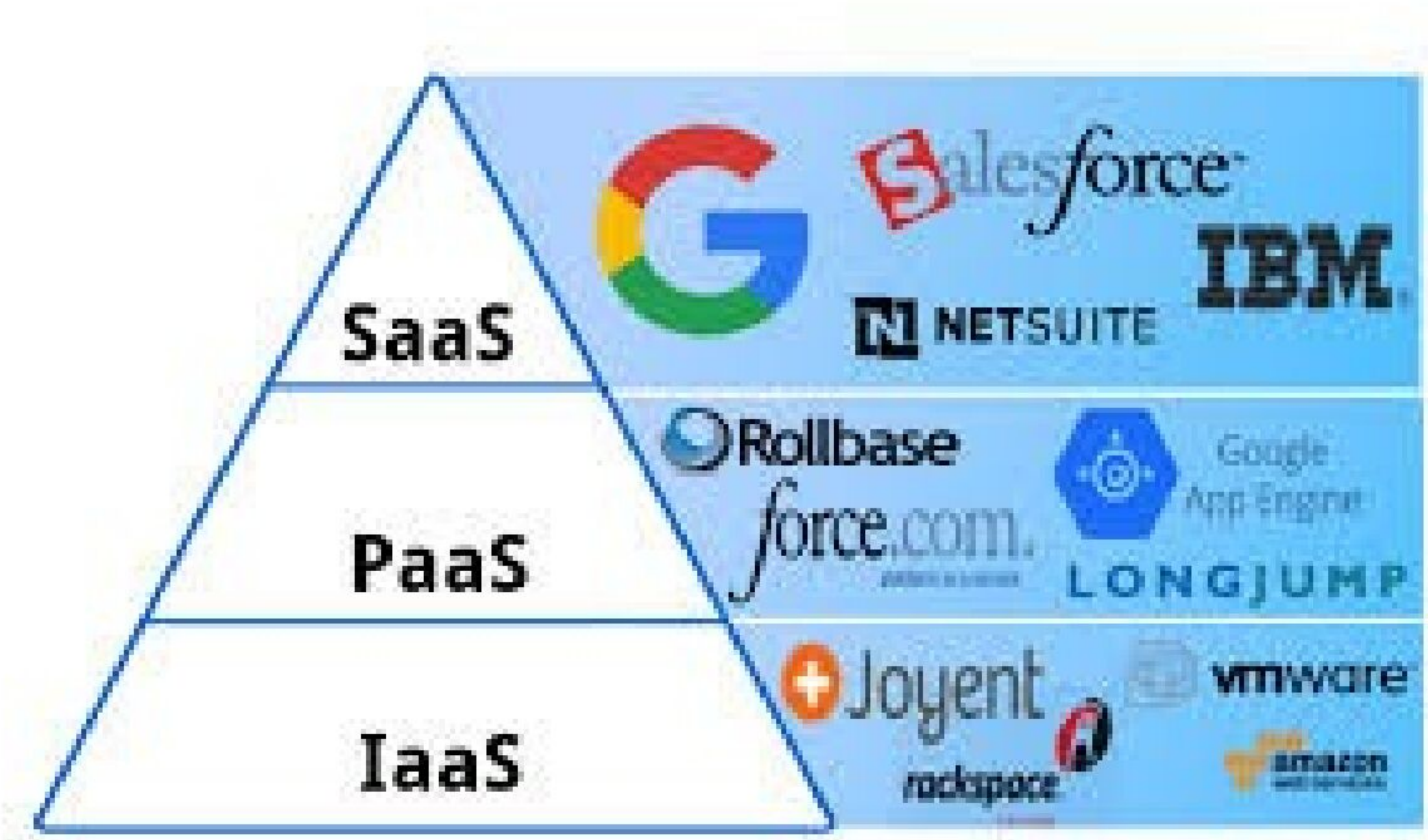
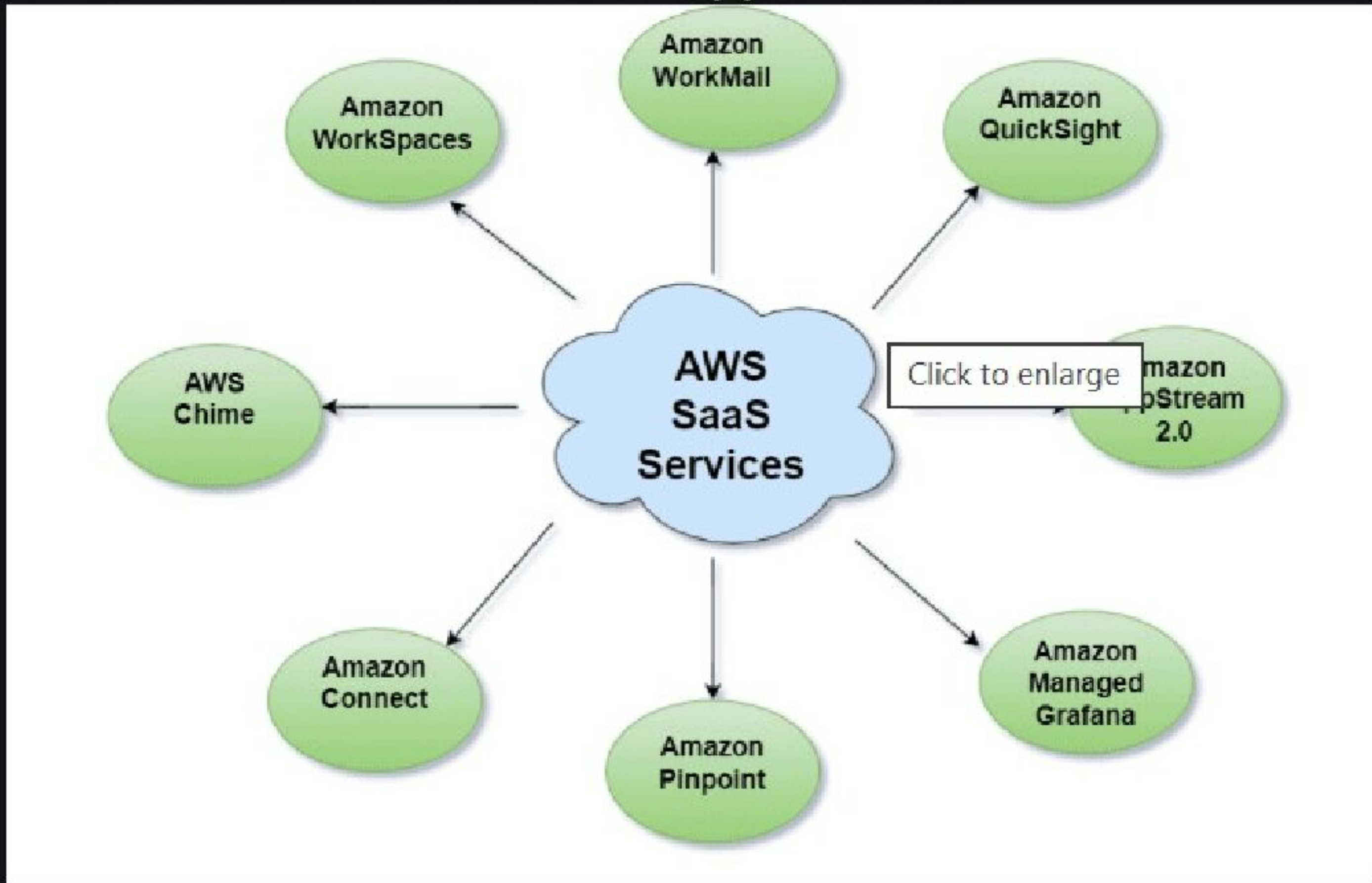
SOFTWARE AS A SERVICE

What is SaaS?

Software as a Service (SaaS) is a cloud-based model where users subscribe to and access applications over the internet via web browsers, rather than installing them locally


SaaS
PaaS
IaaS
salesforce
IBM
NETSUITE
Rollbase
force.com
Google App Engine
LONGJUMP
Joyent
vmware
rackspace
amazon

Top AWS SaaS Solutions for Modern Cloud Applications



AWS SaaS Services

Traditional Software vs SaaS

Traditional Software	SaaS
Installed on local PC	Runs on cloud
High upfront cost	Subscription-based
Manual updates	Automatic updates
Device dependent	Accessible anywhere
Limited scalability	Highly scalable

Software as a Service | SaaS

SaaS is also known as "**On-Demand Software**". It is a software distribution model in which services are hosted by a cloud service provider. These services are available to end-users over the internet so, the end-users do not need to install any software on their devices to access these services.

SaaS Services

Popular SaaS Providers

Key Characteristics of SaaS

- Web browser based access
- Multi-tenant architecture
- Centralized data storage
- Automatic updates
- High availability
- Subscription pricing

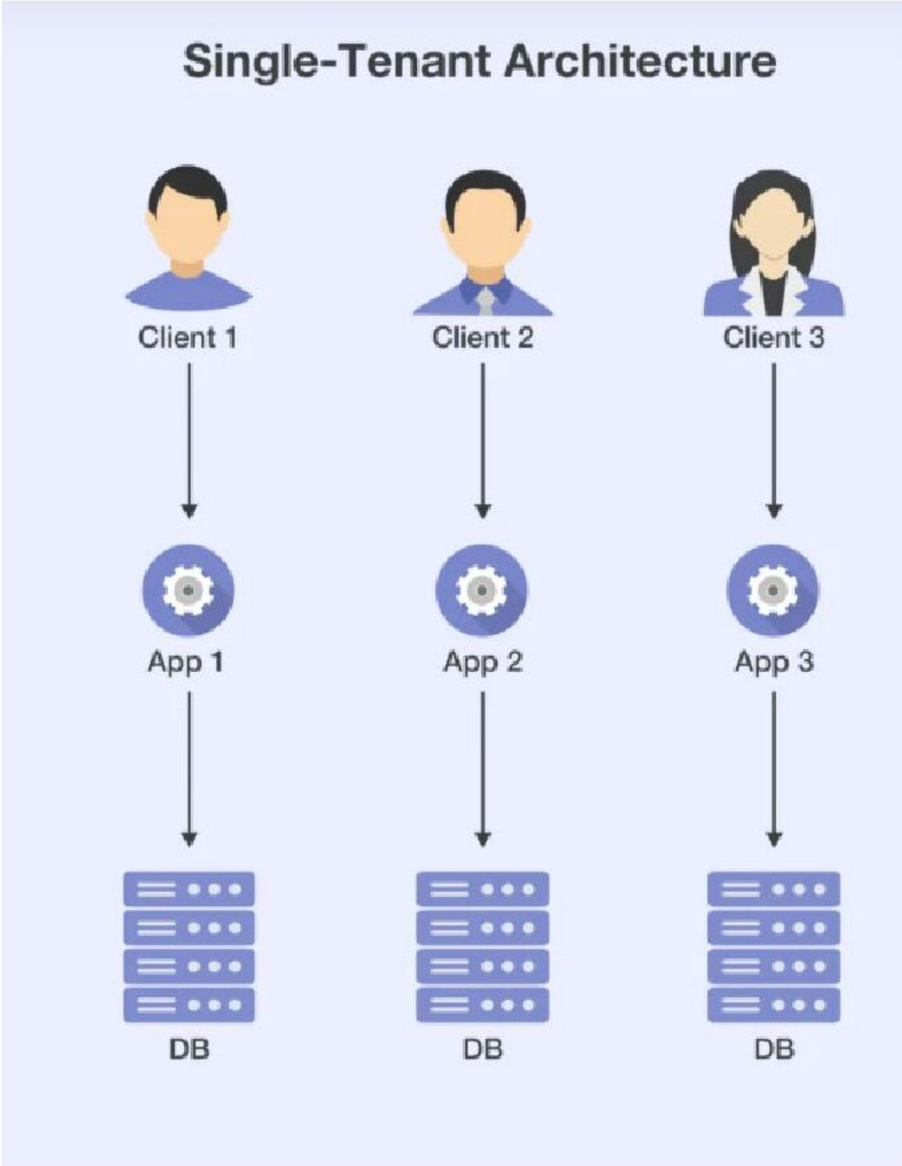
Why SaaS is Popular?

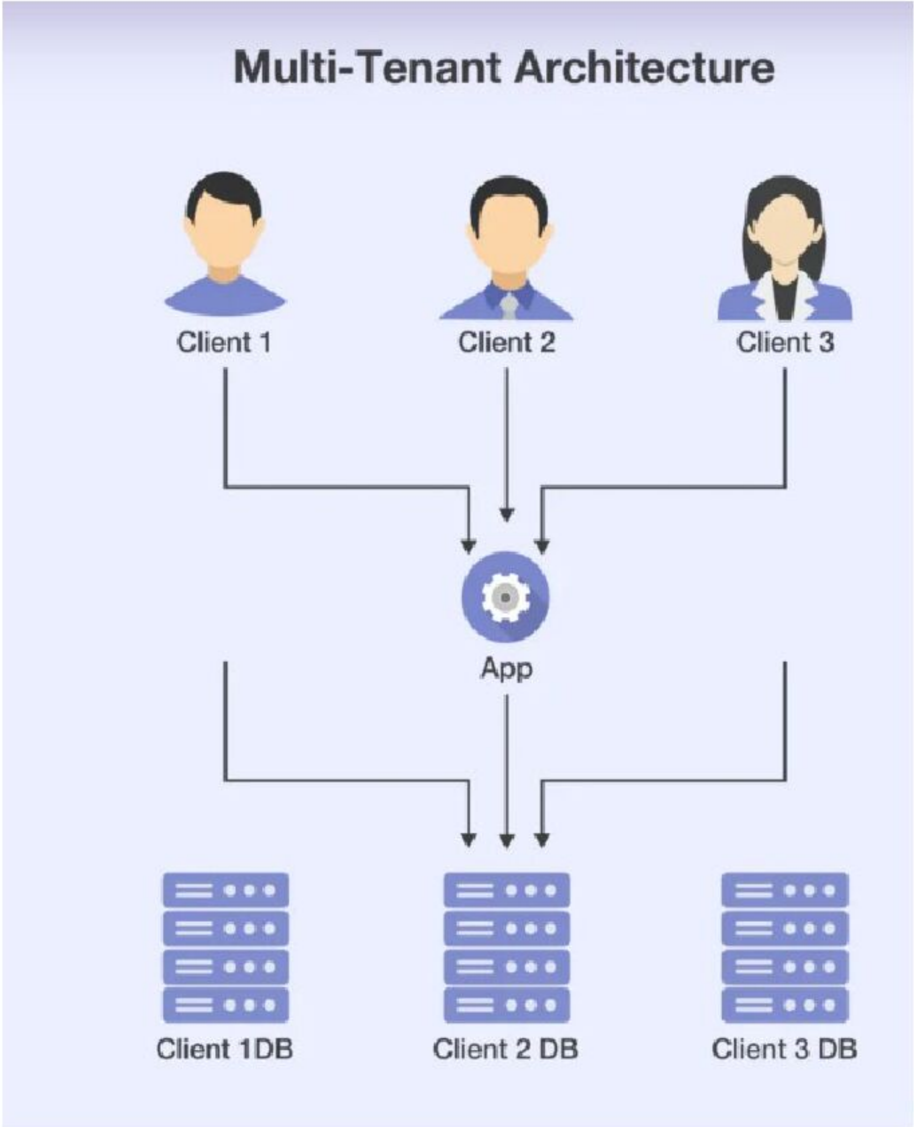
- No installation
- No maintenance
- Lower cost
- Easy collaboration
- Anywhere access

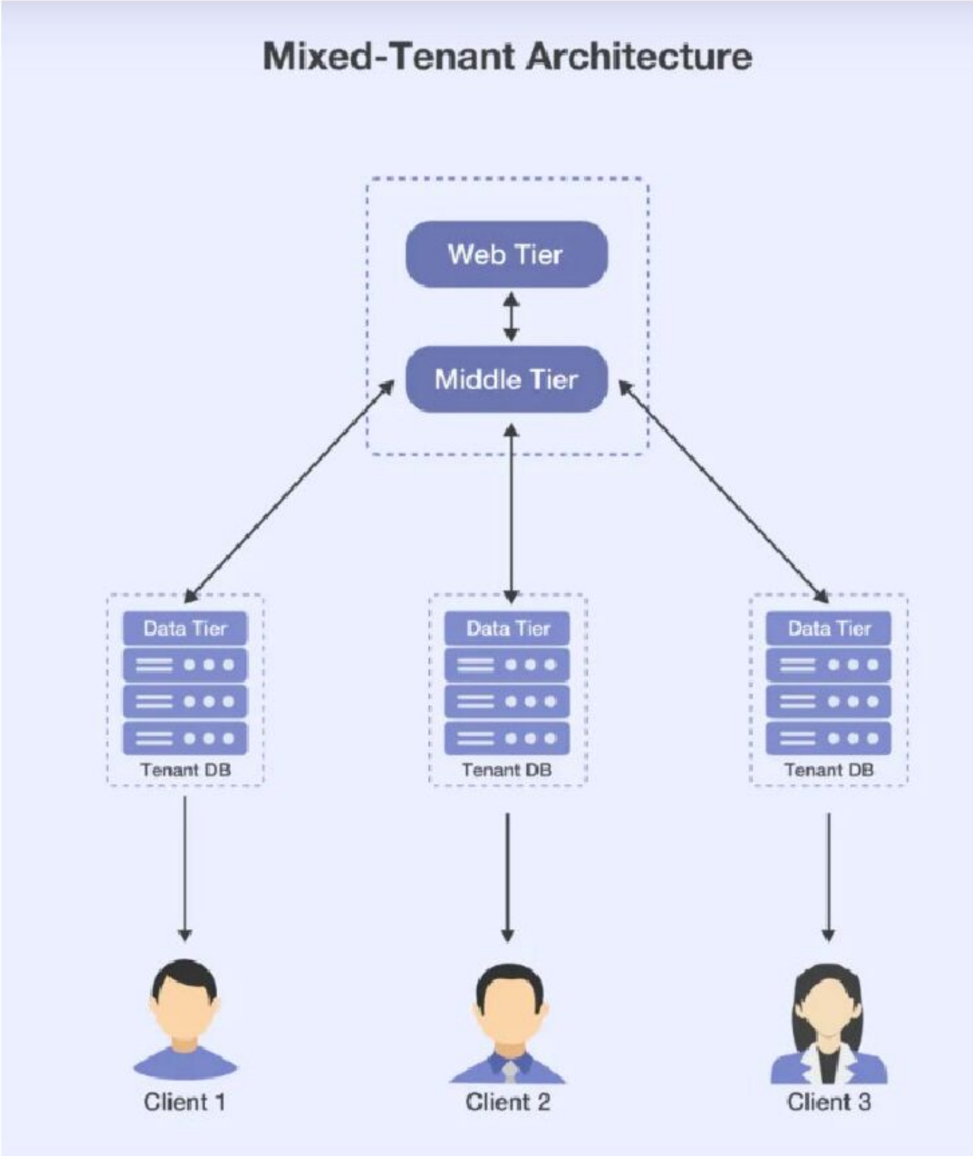
SaaS
ARCHITECTURE & CHARACTERISTICS

SaaS Architecture

User
↓
Web Browser
↓
SaaS Application
↓
Application Logic
↓
Database (Cloud)
↓
Cloud
Infrastructure


Single-Tenant Architecture
Client 1
Client 2
Client 3
App 1
App 2
App 3
DB
DB
DB


Multi-Tenant Architecture
Client 1
Client 2
Client 3
App
Client 1DB
Client 2 DB
Client 3 DB


Mixed-Tenant Architecture
Web Tier
Middle Tier
Data Tier
Tenant DB
Data Tier
Tenant DB
Data Tier
Tenant DB
Client 1
Client 2
Client 3

SaaS Components

User Interface (Browser / App)

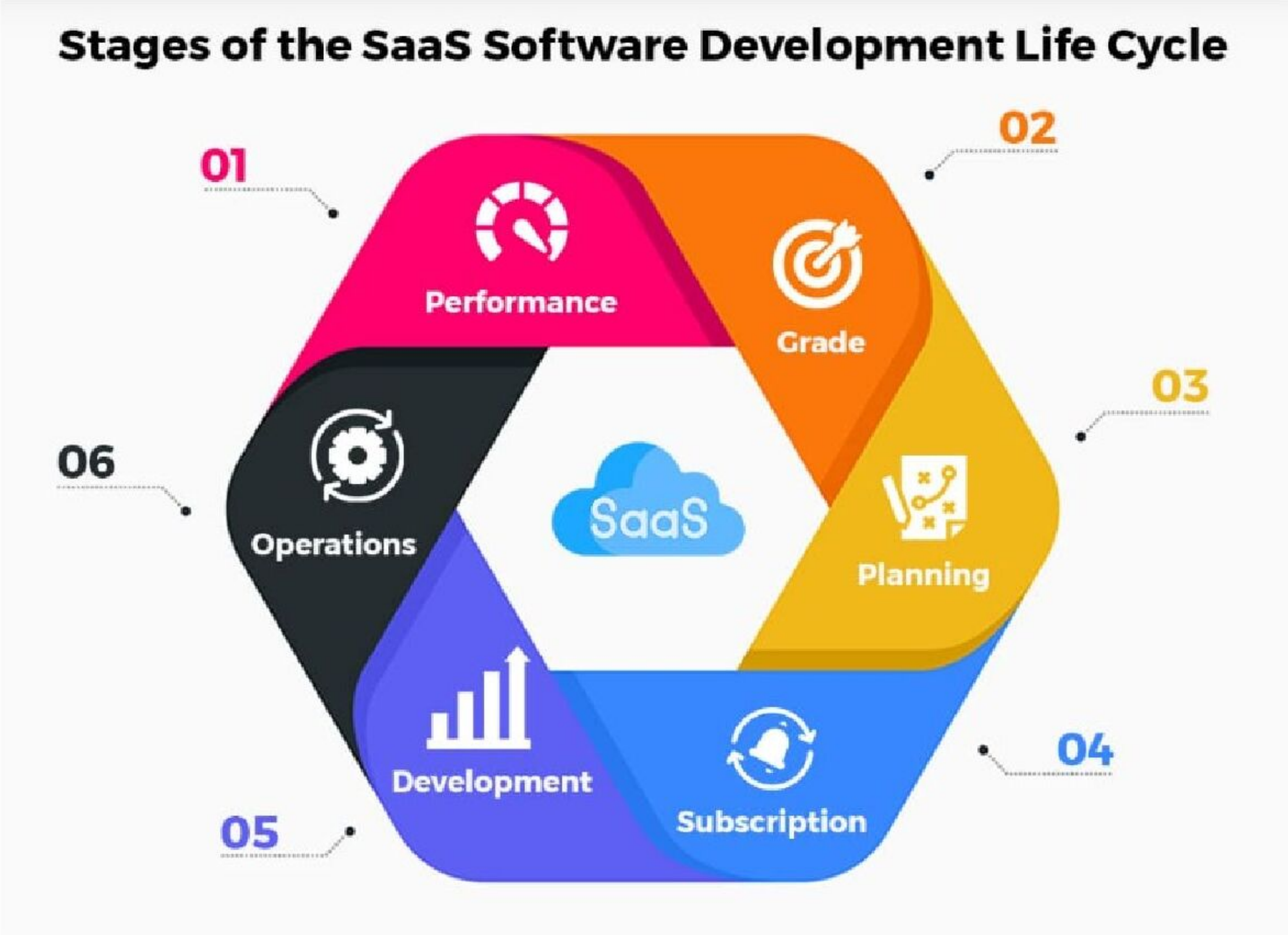
Security & Authentication

Application Logic

Middleware

Database

Cloud Infrastructure


Stages of the SaaS Software Development Life Cycle
01
Performance
02
Grade
03
Planning
04
Subscription
05
Development
06
Operations
SaaS

WEB SERVICES (FOUNDATION OF SaaS)

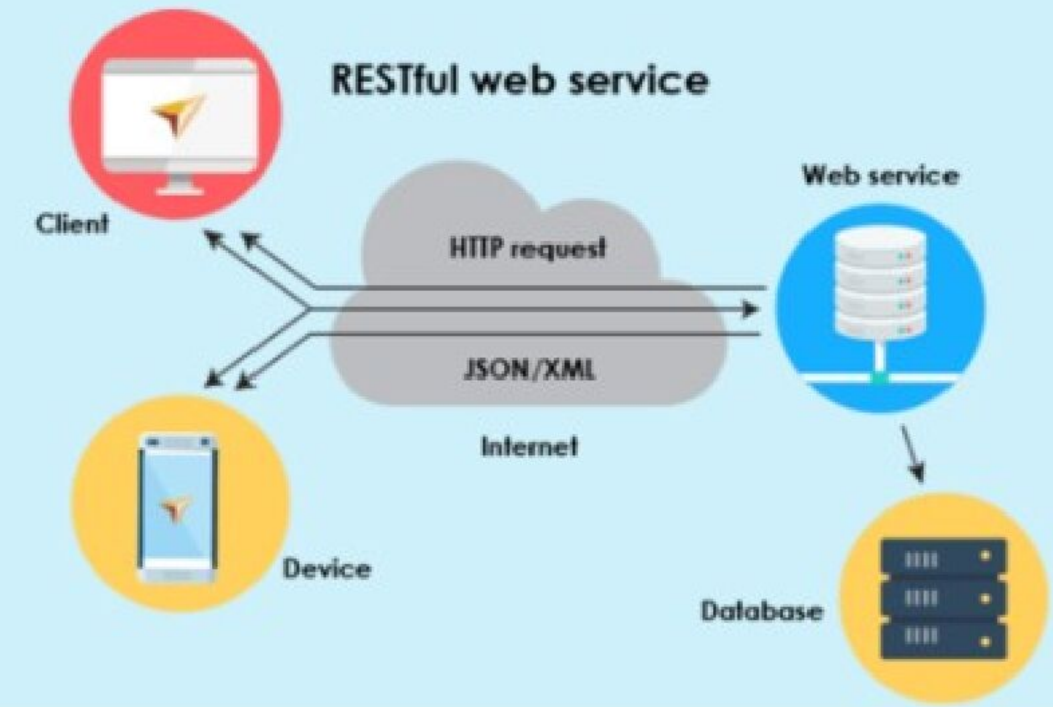
What are Web Services?

Web services are software systems designed to support communication between applications over the web.

SaaS uses web services to:

- Exchange data
- Integrate services
- Enable APIs

Web Service & API Testing



Types of Web Services

1. **SOAP (Simple Object Access Protocol)**

XML-based

Heavy but secure

Used in enterprise apps

2. REST (Representational State Transfer)

Lightweight

Uses HTTP methods (GET, POST, PUT, DELETE)

Most common in SaaS

Role of Web Services in SaaS

Backend communication

Data exchange

Integration with other apps

Mobile app connectivity

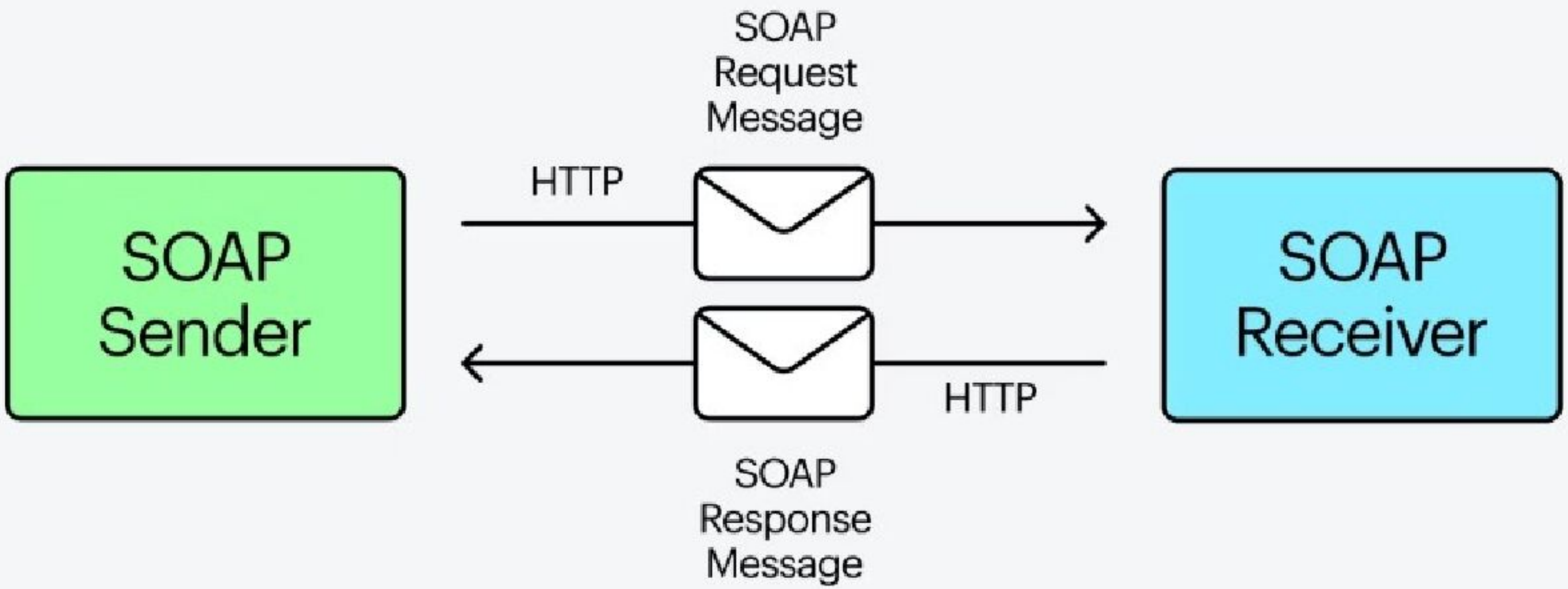
Example:
Gmail uses web services to:

Fetch emails

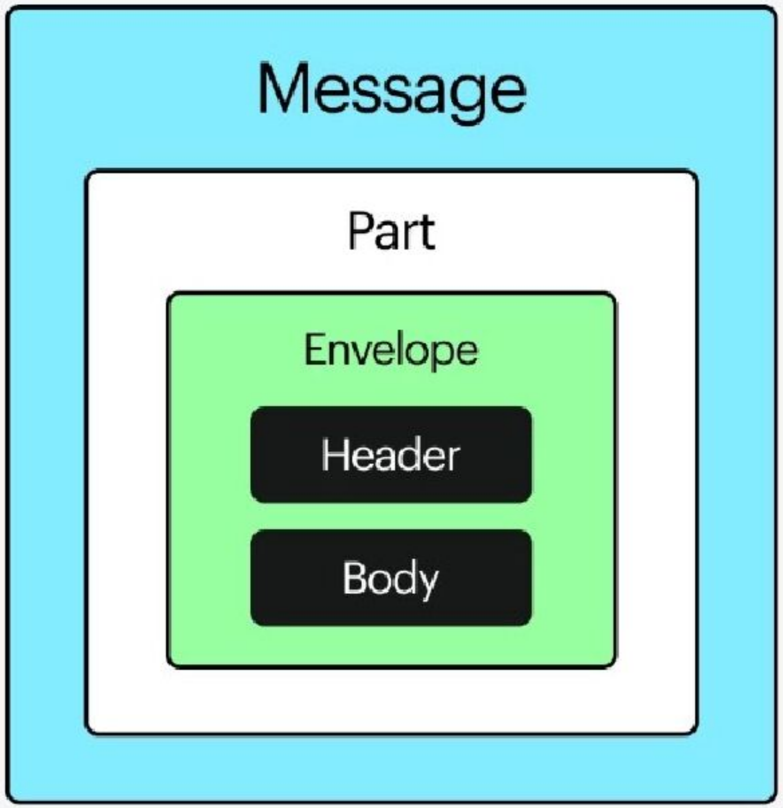
Sync contacts

Send notifications

SOAP (Simple Object Access Protocol) is a messaging protocol used for exchanging structured information between systems over a network, using XML for messaging and HTTP or SMTP for transport.



Components of SOAP



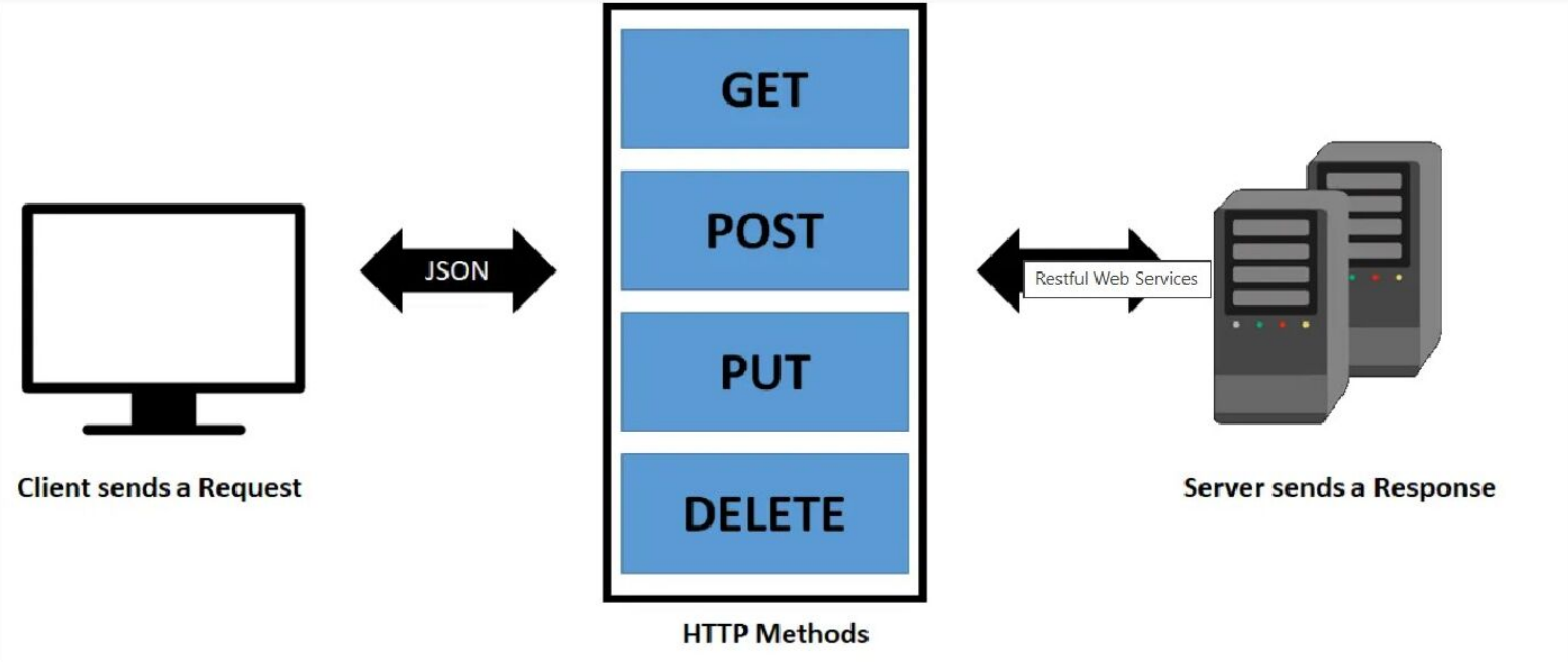
https://www.ramotion.com/blog/soap-in-web-services/

How to Build SOAP Web Service?

1. Determine the requirements of your web service
2. Choose a programming language and framework
3. Define the WSDL (Web Services Description Language) file
4. Implement the web service methods
5. Test and Deploy the web service
6. Maintain and update the web service

What is REST

As stated earlier, REST stands for Representational State Transfer. It is a simple way of sending and receiving data between client and server. It doesn't require any software or standards to transfer data. It has a predefined structure to do the API call. Developers just need to use the predefined way and pass their data as JSON payload.



What is Web 2.0?

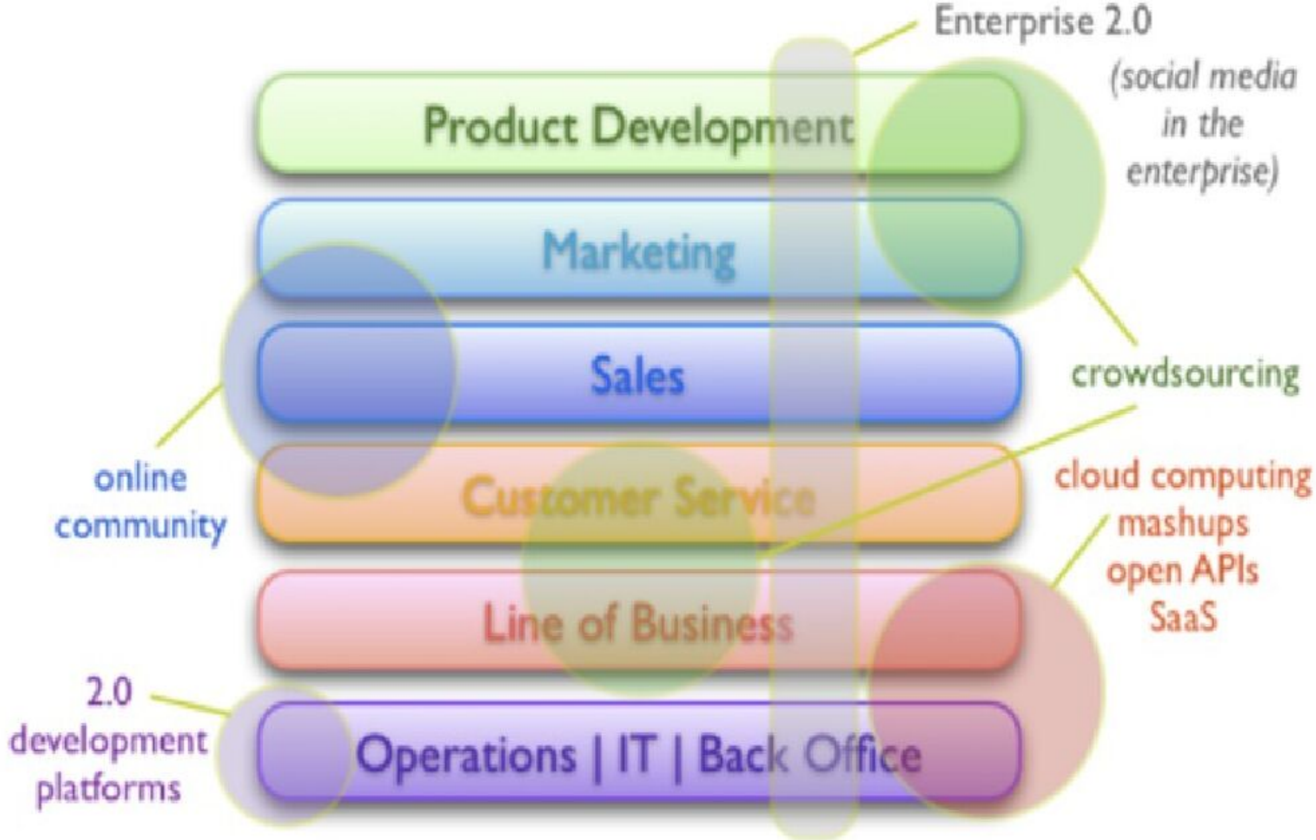
Web 2.0 refers to interactive, user-centric web applications that allow collaboration and content sharing.

Features of Web 2.0	**Web 2.0 Technologies**
User participation	AJAX
Rich user interfaces	JavaScript
Social interaction	HTML5
Dynamic content	CSS3
APIs & mashups	APIs

Relationship Between Web 2.0 and SaaS

Web 2.0	SaaS
Interactive UI	Cloud software
User collaboration	Multi-user access
APIs	Web services
Dynamic content	Real-time updates

Strategic Application of Web 2.0 in Business



From http://blogs.zdnet.com/Hinchcliffe

WEB OS & SaaS

A Web OS is an operating system that runs inside a web browser and uses cloud services.

Web OS vs Traditional OS

Traditional OS	Web OS
Local installation	Browser-based
Hardware dependent	Cloud dependent
Limited collaboration	Built-in collaboration

Features of Web OS

Browser-based interface
Cloud storage
App-centric
Device independent

Role of Web OS in SaaS

Runs SaaS apps smoothly
Reduces hardware dependency
Supports remote work

Examples of Web OS

Chrome OS
eyeOS
Cloud-based desktops

SaaS EXAMPLES & CASE STUDIES

Business:

Salesforce
Zoho CRM

Communication:

Gmail
Zoom

Productivity:

Google Docs
Microsoft 365

Education:

Google Classroom
Coursera

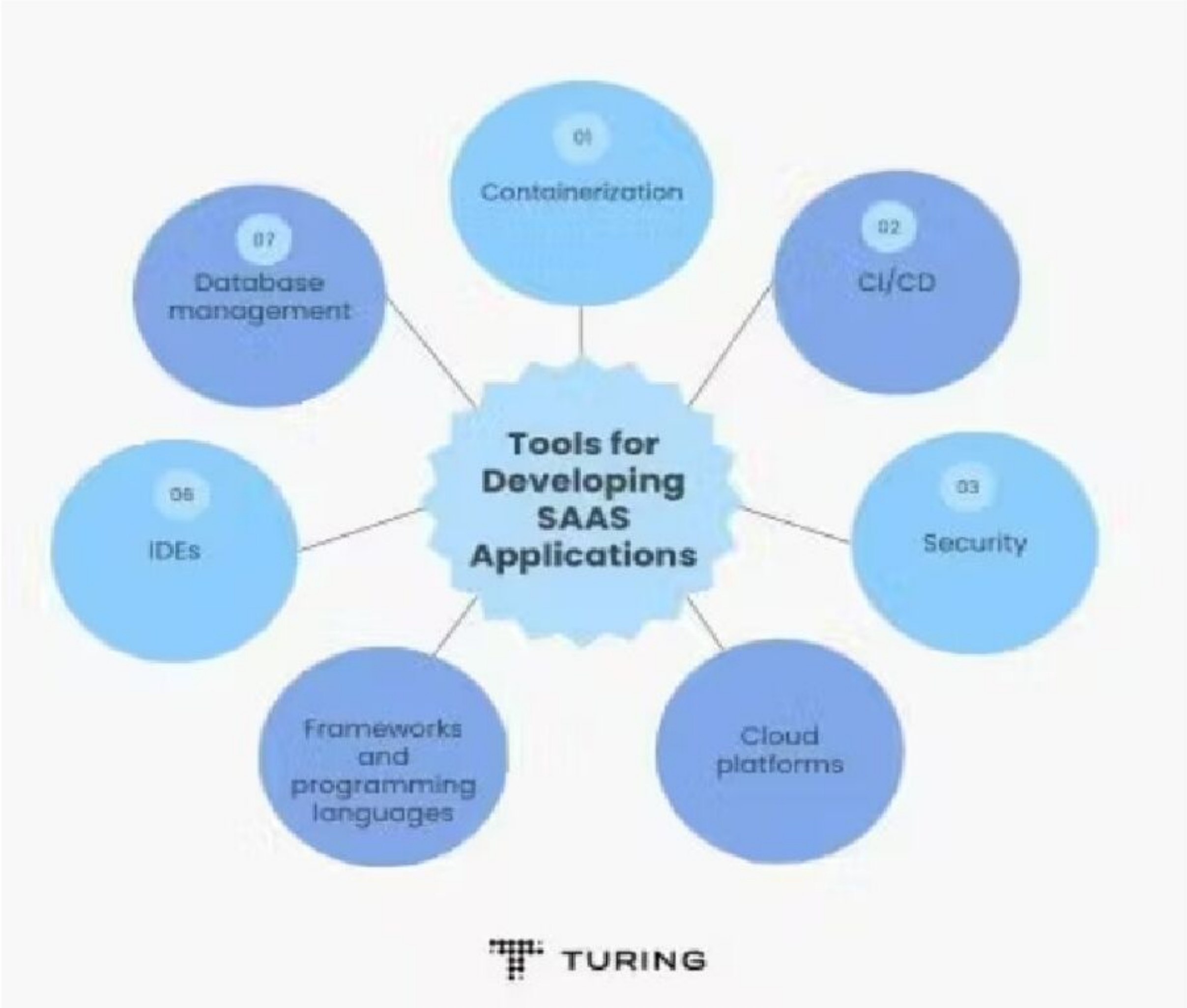
SaaS in Daily Life

Email
Online banking
Video conferencing
Social networking

Case Study: Google Docs

No installation
Real-time collaboration
Auto-save
Cloud storage

HOW TO IMPLEMENT SaaS



Steps to Implement SaaS

1.Identify application

2.Design multi-tenant architecture

3.Develop web-based UI

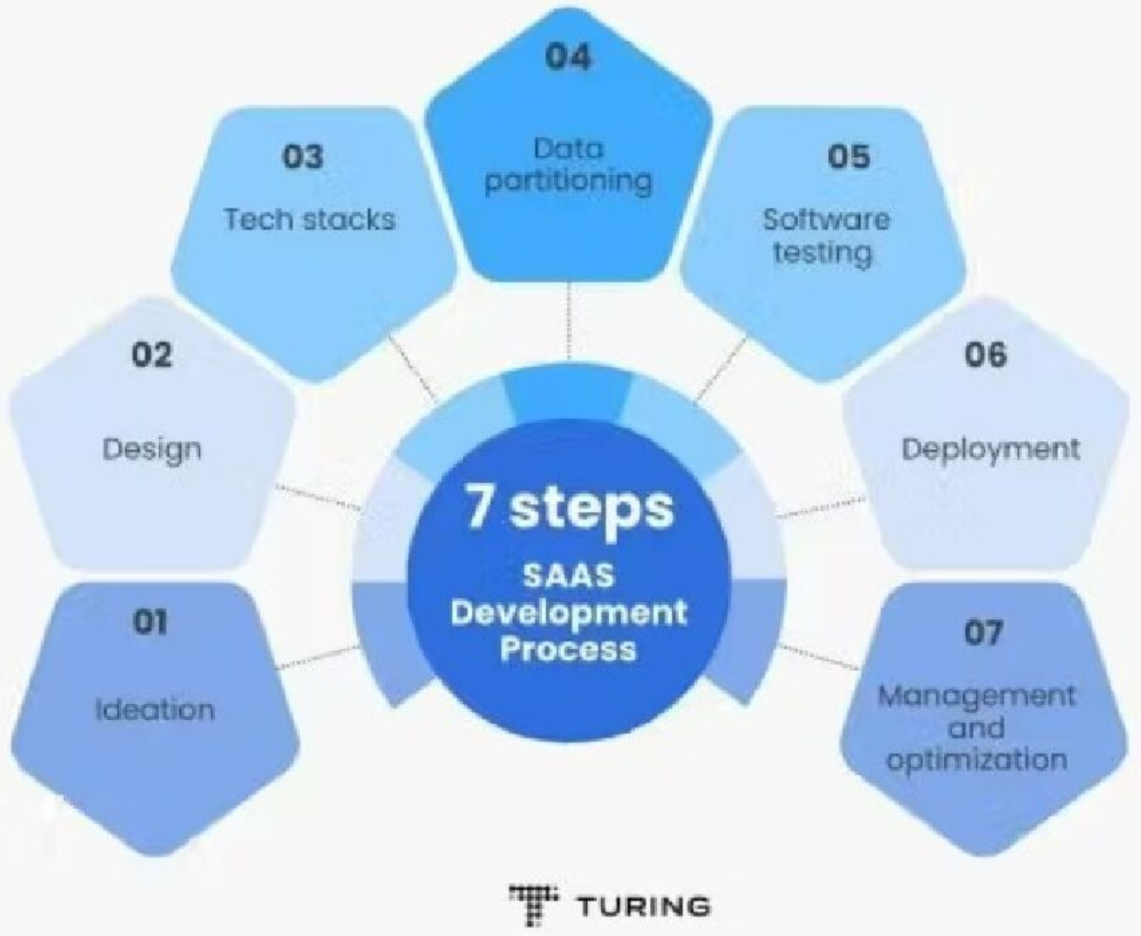
4.Use web services/APIs

5.Deploy on cloud platform

6.Implement authentication

7.Enable billing & monitoring

SaaS development process



User → Browser → SaaS App → Web Services → Cloud DB

SECURITY, ADVANTAGES, LIMITATIONS

User **never installs software**

Access via **URL**

Updates handled by provider

Security in SaaS

•Authentication

•Authorization

•Data encryption

•Secure APIs

•Regular backups

Advantages of SaaS

✔ Cost effective

✔ Easy access

✔ Automatic updates

✔ High availability

✔ Scalability

Limitations of SaaS

- Internet dependency
- Limited customization
- Data security concerns
- Vendor lock-in

	IaaS	PaaS	SaaS
Provider manages...	Infrastructure (network, virtualization, hardware)	Infrastructure and platform	Infrastructure, platform, and software
You manage...	Platform (OS, middleware, runtime) and software (data and apps)	Software	You manage nothing and just use the software.

HubSpot

Characteristics of IaaS, PaaS, and SaaS

IaaS	PaaS	SaaS
Scalability with on-demand resources	Suite of developer tools and services offered	Accessible from any device
Cost-effective: Only pay for the resources used	Middleware services provided	Automatic updates for features and security
High level of customization offered	Automatically scales resources	Subscription-based pricing
Robust security measures such as firewalls, etc.	Easy collaboration on projects	Integration with other cloud services

TURING

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